

ADITYA PANDIARAJAN

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EDUCATION

University of Southern California

Aug 2024 – May 2026

Master of Science in Computer Science; CGPA: 3.6/4

Los Angeles, USA

Coursework: Advance Algorithms, Database Management, Information Retrieval & Web Search Engine

SRM University

Aug 2020 – May 2024

B.Tech in Computer Science and Engineering; CGPA: 9.4/10

Chennai, IN

Coursework: Operating Systems, Computer Architecture, AI & Applications in Cloud, Data Centric Networking

TECHNICAL SKILLS

Languages: Python, Java, C++, C#, Go, TypeScript, JavaScript, SQL

Backend & APIs: FastAPI, Node.js, ASP.NET Core, Spring Boot, REST APIs, WebSockets

Frontend & APIs: React, TypeScript, Tailwind CSS, REST APIs, WebSockets

AI & Agents: OpenAI, Claude, Gemini, LangGraph, LangChain, RAG, Prompt Engineering, Tool Calling

Data & Messaging: PostgreSQL, MongoDB, FAISS, Kafka, Redis Streams, Redis

Cloud & Infrastructure: AWS EC2, Lambda, S3, CloudWatch, Docker, Kubernetes, CI/CD

EXPERIENCE

KVS Technologies

Jul 2025 – Present

Software Engineering Intern, Applied AI Systems

Marietta, GA

- Translated **client demo requirements** into a real-time image-to-insight workflow, enabling a new AI-powered feature through FastAPI, Claude vision models, and SerpAPI-based retrieval.
- Processed **10K+ product workflow** events/day by building a Kafka and Redis Streams event pipeline for user activity tracking, metadata synchronization, and asynchronous AI job execution.
- Reduced **chatbot inference costs** by 30% while stabilizing p95 latency by implementing prompt-level caching, complexity-based model routing, and provider fallback tracing.

USC Information Sciences Institute

Nov 2024 – Jun 2025

Backend Engineering Intern, Agentic Systems

Los Angeles, CA

- Built a LangGraph-based human-in-the-loop workflow for **ad-spend optimization**, reducing risky autonomous budget changes by requiring manager approval before executing AI-recommended shifts.
- Improved **backend throughput** by 47% under high-concurrency workloads by optimizing C# async execution and batching PostgreSQL queries to reduce duplicate reads and database contention.
- Improved **post-launch stability** of backend services by containerizing workloads with Docker, adding Kubernetes health checks and rolling updates, and automating releases through CI/CD pipelines.

Indian Institute of Technology, Roorkee

Jun 2023 – Aug 2023

Software Engineer Intern, AI & Data Systems

Roorkee, India

- Partnered with engineering stakeholders to evaluate **fine-tuning versus prompt-based RAG** for specification queries, selecting hybrid retrieval after comparing citation traceability and hallucination risk.
- Improved **frontend update latency** by building a TypeScript and Tailwind frontend backed by WebSocket streams, enabling real-time updates from backend processing workflows.
- Improved **backend integration reliability** by designing Node.js REST APIs with schema validation, structured data models, and error handling to reduce malformed workflow requests.

SELECTED PROJECTS, OPEN SOURCE AND PUBLICATIONS

Banking Compliance RAG Evaluator | Python, Ragas, LangChain, OpenAI, FAISS

Jan 2026

- Built a Ragas-based evaluation harness for banking policy QA, benchmarking chunk sizes, top-k retrieval, and prompt variants using faithfulness, answer relevancy, context precision, and context recall.
- Created a golden test set of compliance questions to detect hallucinated answers and weak retrieval paths.

Secure-Ops: Multi-Tenant HR & Payroll Agent | FastAPI, LangGraph, PostgreSQL, JWT, Presidio

Jan 2026

- Built a multi-tenant HR/payroll agent with JWT auth, RBAC middleware, and tenant-scoped LangGraph tools to enforce role-based access across salary, benefits, and employee-record workflows.
- Implemented permission guards and PII redaction with role-aware PostgreSQL filters and Presidio.

Publications:

- A Comparative Analysis on Ophthalmology Board-Style Questions. *IEEE Nature Publishing Group UK* 2025.
- Analysing Migraine Patterns Using Ensemble Learning. *IEEE IconDeepCom* 2024.
- BlockFund – Leveraging Escrow and Milestone Algorithm for Secure Crowdfunding. *IEEE ICIMIA* 2023.

HONORS AND AWARDS

Merit-Based Scholarship Recipient, awarded to the top 0.1% of students at the university for outstanding academic excellence combined with notable achievements in sports and extra-curricular.

Kavach All-India Hackathon Runner-Up, developed a real-time license plate detection system to automate parking validation and streamline vehicle-based payment workflows.

AI Tutor Hackathon Second Runner-Up (USC), developed an intelligent tutoring system for all aged-students.